

Supervised Tumor Attention Transfers Across CT Cohorts, but EGFR Prediction Does Not Exceed Clinical Variables

Article type: Original Research

Summary statement. Mask-supervised tumor attention transferred to an external cohort in another country, whereas prediction of epidermal growth factor receptor mutation status from the same images did not exceed a five-variable model of routine clinical data.

Key points

- In 153 patients with a known EGFR call, pooled out-of-fold area under the receiver operating characteristic curve was 0.618 ± 0.025 , compared with 0.794 for smoking status alone; adding the model to a five-variable clinical model changed it by -0.029 ± 0.015 , with every seed's confidence interval including zero.
- Mask-supervised attention transferred to 420 external patients from another country and scanner fleet. Because the crops are lesion-centered, a fixed center prior already reaches 0.270 of 0.292 on attention mass; the informative margins are pointing accuracy 0.609 versus 0.476 and area-matched overlap 0.394 versus 0.287, both invariant to the prior's width.
- Across six defensible preprocessing choices the radiomics comparator moved 0.136 in discrimination, more than the effect sizes this literature reports as findings.

Keywords: CT; Lung; Radiogenomics; Supervised Learning; Technology Assessment.

Abbreviations: AUC = area under the receiver operating characteristic curve, CI = confidence interval, CV = cross-validation, IQR = interquartile range, NSCLC = non-small cell lung cancer, ROI = region of interest, TKI = tyrosine kinase inhibitor.

Abstract

Purpose: To evaluate whether a vision encoder with mask-supervised attention predicts EGFR mutation status from lung CT beyond routine clinical variables, and whether its attention stays aligned to the tumor on an unseen cohort.

Materials and Methods: This retrospective secondary analysis used the public NSCLC-Radiogenomics collection, whose CT dates are de-identification offsets, so no accrual window is recoverable. Of 211 patients, 153 had a known EGFR call (mean age, 69 years $\pm$ 9 [SD]; 98 male, 55 female; 40 mutant [26.1%]). A BiomedCLIP ViT-B/16 encoder with mask-supervised region attention was trained with nested five-fold cross-validation, three seeds. Inputs were lesion-centered crops, so the endpoint measures attention alignment, not detection. That endpoint was attention-mass lift over a per-fold spatial shuffle on 420 held-out NSCLC-Radiomics patients, with a center prior and untrained head as controls. A co-primary endpoint, incremental area under the receiver operating characteristic curve (AUC) over a five-variable clinical model, was pre-declared as an expected null. P values less than .05 indicated significance.

Results: Pooled out-of-fold EGFR AUC was 0.618 ± 0.025 against 0.794 for smoking status alone. Adding the model to clinical variables changed AUC by -0.029 ± 0.015 ; the best fusion reached 0.736 ± 0.005 , below the clinical model. Calibration slope was 0.19, and Brier score exceeded the base-rate floor (0.236 vs 0.193). Attention transferred: mass lift was +0.244 over the shuffle in 11 of 15 evaluations. A center prior reached 0.270 of the model's 0.292 on mass, but pointing accuracy was 0.609 versus 0.476 and overlap 0.394 versus 0.287.

Conclusion: Mask-supervised tumor attention transferred across institutions, whereas lesion-anchored EGFR prediction did not exceed a five-variable clinical model.

Introduction

EGFR genotype directs tyrosine kinase inhibitor (TKI) therapy in non-small cell lung cancer (NSCLC). Tissue is often limited and molecular testing is not universally available, so a non-invasive imaging surrogate would have clinical value, and a large radiogenomics literature has followed.

That literature reports EGFR prediction from CT at area under the receiver operating characteristic curve (AUC) 0.80–0.89 (1–5). The cohorts are typically 200–826 patients, usually single-center, and external validation is uncommon. Against this, the one large study with external validation — 1646 patients and 11 473 segmented lesions with next-generation-sequencing labels — reports internal AUC 0.62–0.68 and external 0.55–0.63 (6). Its external cohort is NSCLC-Radiogenomics, the collection used here.

That discrepancy is this study’s motivation. Reported performance in this field is sensitive to methodologic choices that are invisible at internal evaluation (7, 8), which makes separating the claims more informative than another point estimate. We therefore do not propose a better classifier. We separate two claims that are routinely bundled — that such a model localizes the lesion, and that it genotypes it — and test each with the controls it requires, on a cohort where the second can be measured against the clinical variables it would have to beat.

Materials and Methods

Ethics

This retrospective secondary analysis used publicly available, de-identified imaging and clinical data distributed by The Cancer Imaging Archive under open licenses. No protected health information was accessed.

[AUTHOR ACTION REQUIRED BEFORE SUBMISSION] Insert the institutional review board’s written determination here — formal approval, a waiver of informed consent, or an exempt / not-human-subjects-research determination with its reference number and date — together with the informed-consent statement and, for a United States institution, HIPAA-compliance status. RSNA requires these in the first paragraph of Materials and Methods, and the authors’ reading of the regulations is not a substitute for the board’s determination.

Study design and cohorts

The analysis plan, including the primary endpoint and all controls, was fixed and version-controlled before external evaluation; deviations from it are logged in the public repository accompanying this work.

The internal cohort is NSCLC-Radiogenomics (9), a surgical series distributed by The Cancer Imaging Archive. Participant flow is given in Figure 1: 211 patients in the clinical sheet, 188 with an imaging series, 158 successfully preprocessed, 153 with a known EGFR call, 133 with adenocarcinoma, and 97 with both adenocarcinoma histology and a pixel-level segmentation.

No accrual window is reported: the collection’s CT dates span 1989–1996 and are de-identification offsets, so publishing a range derived from them would misdescribe the cohort. The interval from index test to reference standard, which survives the offset because it is a difference, has a median of 38 days (IQR, 19.5–65.5).

The external cohort is NSCLC-Radiomics (10), 420 preprocessed patients from a Dutch center — another country, scanner fleet, and clinical population. The two cohorts share no patients, and no evaluated checkpoint was warm-started from the external collection; both are asserted in the accompanying test suite rather than claimed in prose.

Reference standard

EGFR status was determined by sequencing and is distributed with the collection (9). The reference standard for attention is the expert tumor segmentation distributed with each collection. Only one contour per lesion exists, so interreader agreement cannot be computed; a

mask perturbation sweep is reported instead and is not a substitute.

Model

The image encoder is the vision tower of BiomedCLIP (ViT-B/16, last two blocks unfrozen), producing patch tokens per section, aggregated across 16 sections by a masked mean. A region head applies eight learned queries by cross-attention, and the resulting maps are supervised against the tumor region of interest (ROI) projected onto the patch grid. The classifier pools from those region embeddings, so prediction and attention share a representation. A language decoder exists in the codebase but its loss weight is zero in every run reported here and its output does not reach the classifier; this is an image-only model.

The attention task is ROI-conditioned, not whole-thorax detection. Inputs are axial crops of $(\text{bbox}_{\text{mm}} + 24) \times 2.5$ around the lesion with 30% positional jitter, windowed at center -600 HU and width 1500 HU, and resized to 224×224 pixels. The lesion sits near the center of the field of view by construction, and the model is never asked to find a tumor in a whole chest CT. A center prior is therefore a mandatory control, and the metrics invariant to its width carry the argument. Because the attention maps are supervised directly against the reference segmentation, we call this mask-supervised attention rather than weakly supervised localization, which denotes attention emerging from image-level labels alone.

Endpoints and controls

H_1 (primary): external attention-mass lift over a per-fold spatial shuffle is greater than zero.

Three controls appear in every reported table:

- Spatial shuffle — the same attention values permuted across cells, isolating where mass fell from how peaked the map is.
- Center prior — external ROI centroids sit at $(0.50, 0.51)$ of the frame covering 4.9% of the area, so “look at the middle” is a strong baseline. Pointing accuracy and area-matched intersection over union are invariant to the prior’s width, so that margin cannot be tuned.

- Weight randomization — an untrained head of identical architecture, ruling out architectural inductive bias.

H₂ (co-primary, pre-declared null): $\Delta\text{AUC} = \text{AUC}(\text{clinical} + \text{model}) - \text{AUC}(\text{clinical})$, where the combination is an unweighted average of within-fold ranks, that is, soft voting. This topology was pre-specified; the alternatives in Table 2 are a reported ablation and not the endpoint.

The five-variable clinical model is a logistic regression on age, sex, self-reported ethnicity, smoking status and pack-years. Numeric variables are median-filled with a missingness indicator and categorical variables get an explicit unrecorded level, so no patient is dropped and every arm is scored on the same patients. Imputation, encoding and tuning occur inside the training folds.

Endpoints and their populations. H₁ uses the 420 external patients; H₂ and every arm in Table 2 use the 153 internal patients with a known EGFR call. The 97-patient segmented-adenocarcinoma set is the pre-specified analysis set for the histology-restricted sensitivity analysis below, not for the headline figure.

Statistical analysis

Evaluation used nested five-fold cross-validation (CV) stratified on EGFR, three seeds, with within-fold rank normalization before pooling. The headline is the across-seed mean $\pm$ SD; a pooled confidence interval (CI) absorbs no rerun variance and is never quoted alone. No operating point is defined, because the calibration reported below does not support a threshold, so sensitivity and specificity are not reported.

Comparisons used a paired bootstrap over patients (2000 resamples), with Holm correction across the ten arms of the pooling sweep. Precision was computed by the Hanley–McNeil closed form (11), validated against a bootstrap; minimum sample size follows Riley et al (12), applied only where its inputs are defined. Reporting follows the Checklist for Artificial Intelligence in Medical Imaging (13); the completed checklist accompanies this manuscript.

P values less than .05 indicated statistical significance, and all tests were two-sided. Analyses were performed in Python 3.13.9 (Python Software Foundation) with NumPy 2.1.3, SciPy 1.15.3,

scikit-learn 1.6.1, pandas 2.2.3 and statsmodels 0.14.4 (open-source, respective maintainers), and PyTorch 2.10.0 (Meta Platforms).

Use of artificial intelligence tools

[AUTHOR ACTION REQUIRED BEFORE SUBMISSION] Generative AI assistance was used in this work and must be disclosed here as well as in the cover letter, naming the tool, version, manufacturer and dates of access, and stating precisely what it was used for — writing and refactoring analysis code, drafting and editing text, literature search, or figure generation. State that all analytical decisions, the pre-specified analysis plan, the interpretation of results and the final text were reviewed and approved by the authors, who take responsibility for the integrity and accuracy of the work, and that no AI tool is listed as an author. Verify the exact model version and access dates against the repository history before submitting.

Results

Patient characteristics

Of 211 patients, 153 had a known EGFR call and formed the analyzed sample (mean age, 69 years $\pm$ 9 [SD]; 98 male, 55 female), of whom 40 (26.1%) were mutant. Never-smokers were 20 of 40 (50%) of mutants against 13 of 113 (12%) of wild-type patients, and 133 of 153 (87%) had adenocarcinoma (Table 1).

The model does not exceed the clinical variables

Table 2 gives every arm on identical patients and folds. Pooled out-of-fold EGFR AUC was 0.618 ± 0.025 across three seeds. Smoking status alone reached 0.794 and the five-variable clinical model 0.764.

The co-primary endpoint was $\Delta\text{AUC} = -0.029 \pm 0.015$, with every seed’s CI including zero. No fusion topology exceeded the clinical model: soft voting of the model and clinical arms reached 0.736 ± 0.005 , cross-fitted stacking 0.718 ± 0.014 , and hard voting 0.688 ± 0.006 .

Radiomics helped neither arm, and the direction depends on which it is added to: added to the

clinical model it reduced AUC from 0.764 to 0.703 (-0.061 ; $P = .02$ in all three seeds), while added to the imaging model it changed nothing (0.618 to 0.619 ± 0.015 ; $P = .87$). A logistic regression on frozen encoder features scored 0.617 ± 0.052 , indistinguishable from the full network to three decimals.

Results are shown per arm in Figure 2 and with paired differences in Figure 3. KRAS was at chance (0.489 ± 0.005); Gevaert et al likewise report no significant KRAS model on this cohort (14).

The predictions are also uncalibrated

Calibration was assessed on the raw per-patient out-of-fold probabilities. The within-fold rank normalization used for pooling discrimination is applied afterwards and never enters these figures, because rank-normalized values are not probabilities.

Calibration slope averaged 0.19 against a perfect 1.0, indicating predictions roughly five times too extreme. At a base rate of 26.1%, a model predicting the prevalence for every patient achieves a Brier score of 0.193; the model scored 0.236. The probabilities are therefore worse than a constant, with resolution — the component only a better model can improve — between 0.009 and 0.017. This is why no operating point is pre-specified, and why any future externally validated artifact must carry explicit recalibration.

On the subset that should carry it, performance is weaker

Every EGFR-mutant patient had adenocarcinoma; the 17 with squamous cell carcinoma and three with NSCLC not otherwise specified were wild-type without exception, making 20 of 153 (13%) of the cohort classifiable from histology alone. On the segmented-adenocarcinoma set ($n = 97$, 23 mutant) AUC falls to 0.563. The same pattern appears in the literature: Yamazaki et al report 0.774 across histologies but 0.687 restricted to adenocarcinoma (15).

Localization generalizes to an unseen cohort

On 420 held-out NSCLC-Radiomics patients (Figure 5), attention mass inside the tumor was 0.292 against 0.048 for the spatial shuffle — a lift of +0.244, exceeding the shuffle in 11 of 15 evaluations, which meets H_1 as pre-specified.

The shuffle is the weak control, and we do not rest the claim on it. A Gaussian blob fixed at the frame center scores 0.270 on mass against the trained model’s 0.292, so most of the margin over a shuffle is reproduced by looking at the middle of a lesion-centered crop. The informative comparison is against that center prior on the metrics it cannot exploit: pointing accuracy 0.609 versus 0.476 and area-matched intersection over union 0.394 versus 0.287, both invariant to the prior’s width. An untrained head of identical architecture returned a mass lift of exactly +0.000 at a peak-to-mean ratio of 1.01 and pointing accuracy 0.005, ruling out architectural inductive bias.

The fold count is a weak estimator, reported because it was pre-specified. The margin over the threshold is one fold, 11 of 15 (73.3%) against 71.4%, and the 15 evaluations are not 15 studies: they are three seeds by five folds scored on the same 420 patients with overlapping training sets, so the count treats dependent replicates as independent. A patient-level analysis resampling the 420 patients is reported below as post hoc. We did not substitute it for H_1 , because choosing a better endpoint after seeing the data is the failure pre-specification exists to prevent.

Table 3 reports every evaluation individually, and the distribution is bimodal rather than marginal: the weakest success has pointing accuracy 0.757 and the strongest failure exactly 0.000, with nothing between. Folds either align decisively or not at all, so the verdict is insensitive to where the threshold sits within a wide interval while remaining sensitive to the fold count.

The failures are concentrated rather than scattered: one run aligned in two of five folds while the others reached five of five and four of five, and that same run aligns in two of five folds internally while carrying the highest classification AUC of the three. Attention alignment is therefore training-unstable, and nothing in the internal metrics predicts which run will transfer.

Post hoc: the margin over the center prior depends on treating the training run as fixed.

Resampling the 420 external patients, mean pointing accuracy exceeded the center prior by

+0.133 (95% CI: 0.082, 0.183; $P < .001$) and area-matched overlap by +0.107 (95% CI: 0.082, 0.133; $P < .001$). Resampling the three training runs as well, both intervals include zero (+0.133; 95% CI: -0.128, 0.350; $P = .30$, and +0.107; 95% CI: -0.064, 0.248; $P = .19$). Only the contrast against an untrained head survives that reading (mass +0.243; 95% CI: 0.130, 0.351; $P < .001$).

Between-run variance therefore dominates — the instability reported above seen from a second direction. A hierarchical interval on three runs is wide by construction and underpowered at the run level, but that is the finding rather than an excuse for it: we can claim these models attend to the lesion on an unseen cohort, not that the next one would.

Representative attention maps appear in Figure 6.

The peritumoral region carries the signal, but not significantly

Because the classifier pools from the attended region — the lesion itself — we asked whether the EGFR signal sits where the attention loss supervises. Over ten seeds on frozen encoder features, mean-pooling the peritumoral ring scored 0.695 against 0.632 for the whole crop and 0.629 for the tumor interior; a size- and shape-matched ring at a random location scored 0.575, so the gain is location-specific. Adding a peritumoral branch moved the trained model to 0.675 ± 0.017 .

This does not survive correction and we report it as suggestive: across the ten arms of the sweep the best has $P_{\text{Holm}} = .28$, and end to end the branch is worth +0.058 (95% CI: -0.008, 0.128; $P = .09$). It replicates six prior reports at ring widths from 1 to 15 mm (1–5, 15). What is new is the tension: the attention loss supervises onto the lesion, and the classification signal appears to sit just outside it.

Analytic choice is worth more than the effect

On identical features and estimator, the radiomics arm moved 0.136 in AUC across six defensible preprocessing choices, and 0.066 between a flat and a nested CV protocol — both larger than the effects this literature reports as findings, and both invisible to internal evaluation (7).

More patients would not settle it

Riley’s criterion requires 414 patients for the five-variable clinical model alone and 42 362 for the 512-feature frozen probe; this cohort has 153. Neither arm is identifiable at this sample size, which is why the frozen probe matching the model cannot be read as evidence that the architecture is redundant.

Precision, not power, binds any external test. A 95% CI half-width of 0.05 requires $n = 661$ at the observed prevalence, while the entire open world of EGFR-labeled CT holds roughly 327 cases with 64 positives, giving 0.154 — an interval spanning chance and the published 0.80s at once.

Discussion

Two claims usually bundled together separate cleanly here, and separate differently.

Mask-supervised attention stayed on the lesion in a cohort from another country, scanner fleet and treatment setting. Genotype prediction did not clear a five-variable clinical model on the cohort it was trained on ($\Delta\text{AUC} = -0.029 \pm 0.015$), was not rescued by any fusion topology, and was worse than a constant on the probability scale (Brier score 0.236 against a base-rate floor of 0.193).

Two limits bound what “transferred” means. The attention task is ROI-conditioned — the lesion is near the center of the input by construction, so this is alignment, not detection in a whole chest CT — and the transfer is training-unstable. The post-hoc analysis makes the second limit precise: the margin over a center prior is unambiguous across patients and disappears once the training run is treated as a random effect. The dissociation below is a statement about the models we trained, not a guarantee about the next one.

What the localization endpoint does and does not establish. Attention mass, pointing accuracy and area-matched intersection over union are not measures of diagnostic accuracy, and no clinical claim follows from them directly. They answer a narrower question that has been assumed rather than tested: whether the explanation a model offers for its prediction is anchored to the lesion at all, on data from an institution it has never seen. That question matters precisely because the accompanying prediction fails. A saliency map that tracks the tumor on an unseen cohort, while

the classifier built on it performs at 0.618, is direct evidence that plausible localization is not evidence of predictive validity — a dissociation that cannot be observed in a study where both are reported as successes. We therefore present H_1 as a methodologic result about interpretability claims, not as a clinical endpoint, and we do not propose the localizer for clinical use.

Convergence with the largest external study. Rodríguez Sánchez et al (6) provide the strongest argument that this is a property of the problem rather than of our sample. Their 1646 patients and 11 473 lesions, with next-generation-sequencing labels and biopsy-anchored training, reproduce the interval we obtain from 153 patients, and their external cohort is the collection we use internally. Ten times the patients did not move the ceiling. Read together with the measured precision bound above — a half-width of 0.154 on the largest test set the open literature can currently assemble — the implication is that the field cannot resolve this question with public data alone.

We are more cautious about what explains the reported 0.80s. Preprocessing choices moved discrimination by 0.136 on identical features here, so analytic flexibility is sufficient to produce such a gap — but it does not follow that it is the explanation in any particular study, and cohort composition, histologic mix, label provenance, scanner differences and undetected leakage could each contribute. The defensible claim is narrower: a reported AUC in this literature is not interpretable without the preprocessing and validation protocol that produced it.

Comparison with prior reports. Our peritumoral finding replicates six prior reports in direction while failing to reach significance after correction for the ten arms actually examined. We consider that a more likely description of this literature than the alternative, given that the same six reports place the optimal ring width anywhere from 1 to 15 mm. Similarly, our histology result reproduces Yamazaki et al (15): performance falls when the patients with squamous cell carcinoma, who are wild-type without exception, are removed. Any study reporting EGFR AUC across mixed histologies is partly reporting a histology classifier.

Limitations. The internal cohort is single-institution and historical, with no recoverable accrual window. The classification arm has no external validation; that is justified by the measured precision bound rather than by circumstance. Interreader variability was not measured, because

the source collections distribute one contour per lesion; a mask perturbation sweep is reported instead and is not a substitute for a second reader. Acquisition parameters are incompletely recoverable. One of three training runs largely fails to align, and we report the endpoint across all three. Finally, H_1 is confirmatory rather than a blind prospective test: the external cohort was on disk, and the metric’s behavior on it partly known, when the threshold was set. We state this rather than let a reader infer a stronger design than was run.

In conclusion, on this cohort mask-supervised tumor attention transferred across institutions while EGFR prediction from lesion-anchored CT did not exceed routinely available clinical variables. Reports of AUC 0.80–0.89 should be read against the analytic-choice sensitivity and precision bounds measured here.

Acknowledgments

[Anonymized for review. Unanonymized acknowledgments appear on the full title page.]

Data and code availability. All code, the pre-specified analysis plan with its deviation log, the completed CLAIM checklist, and every archived run result — including negative results — are available at an anonymized repository for review ([ANONYMIZED REPOSITORY LINK — see cover letter]); the permanent public URL and archival DOI will be provided on acceptance. Every number in this manuscript is regenerated from archived out-of-fold predictions by a single script, requiring no imaging data and no GPU.

Imaging is redistributed by The Cancer Imaging Archive under its own terms and is not included in the repository. NSCLC-Radiogenomics is licensed CC BY 3.0 and NSCLC-Radiomics CC BY-NC 3.0; Figure 6 shows NSCLC-Radiogenomics images only, consistent with those licenses.

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Figure Legends

Figure 1: Flowchart shows participant flow through the study. Counts and exclusion reasons are generated from the analysis pipeline rather than transcribed, and reconcile exactly ($211 - 114 = 97$). The pre-specified primary analysis set is patients with segmented adenocarcinoma. EGFR = epidermal growth factor receptor, NSCLC = non-small cell lung cancer.

Figure 2: Graph shows pooled out-of-fold receiver operating characteristic curves by arm, computed on identical patients and folds. The chance line is drawn and labeled, and each arm's number of patients is printed because the arms do not all cover the same patients. AUC = area under the receiver operating characteristic curve.

Figure 3: Forest plot shows every arm with its 95% confidence interval. Overlapping intervals can still differ significantly when the comparison is paired, which is why the paired differences are reported alongside. AUC = area under the receiver operating characteristic curve, CI = confidence interval.

Figure 4: Graph shows per-seed area under the receiver operating characteristic curve against the clinical baseline. The rerun spread is shown next to the effect being claimed; the gap to a five-variable clinical model exceeds it. AUC = area under the receiver operating characteristic curve.

Figure 5: Graphs show the primary endpoint — localization on 420 held-out NSCLC-Radiomics patients — with all three controls in every panel. The center prior is strong on attention mass by construction and weak on pointing accuracy and area-matched intersection over union, which are invariant to its width, so the trained model's margin is read from the right-hand panels rather than the left. IoU = intersection over union, NSCLC = non-small cell lung cancer.

Figure 6: CT images with superimposed attention maps (hot colors) and the tumor contour drawn from the reference segmentation (green) in six patients: (A) 68-year-old man with adenocarcinoma, (B) 76-year-old man with NSCLC not otherwise specified, (C) 64-year-old man with NSCLC not otherwise specified, (D) 63-year-old man with adenocarcinoma, (E) 84-year-old

man with adenocarcinoma, and (F) 59-year-old man with squamous cell carcinoma. These are the first six patients with a segmentation, in index order, from held-out fold 0 of training run B in Table 3; they were not selected on score, and the histologic mix is whatever that rule produced. Images are from the NSCLC-Radiogenomics collection, used under its CC BY 3.0 license (9). NSCLC = non-small cell lung cancer.

Tables

Table 1: Characteristics of the 153 Patients with a Known EGFR Call

Characteristic	EGFR mutant	Wild-type	All
No. of patients	40	113	153
Age (y)*	—	—	69 ± 9
Age (y)†	68 (63, 76)	69 (64, 75)	69 (64, 76)
Pack-years†	15 (9, 34)	42 (26, 56)	40 (20, 50)
Female	22 (55)	33 (29)	55 (36)
Male	18 (45)	80 (71)	98 (64)
Never-smoker	20 (50)	13 (12)	33 (22)
Former smoker	18 (45)	78 (69)	96 (63)
Current smoker	2 (5)	22 (19)	24 (16)
Asian	9 (22)	6 (5)	15 (10)
White	13 (32)	77 (68)	90 (59)
Other or not recorded	18 (45)	30 (27)	48 (31)
Adenocarcinoma	40 (100)	93 (82)	133 (87)
Squamous cell carcinoma	0 (0)	17 (15)	17 (11)
NSCLC not otherwise specified	0 (0)	3 (3)	3 (2)

Note.—Unless otherwise indicated, data are numbers of patients, and data in parentheses are percentages. Percentages are column percentages within each EGFR group. EGFR = epidermal growth factor receptor, IQR = interquartile range, NSCLC = non-small cell lung cancer.

* Data are means ± SDs. Per-group means are omitted to avoid implying a comparison that was not tested.

† Data are medians, and data in parentheses are IQRs. Both a mean and a median are given for age because the distribution is mildly left-skewed.

Table 2: Discrimination of Every Arm and Fusion Topology on Identical Patients and Folds

Arm	AUC	Difference vs clinical	P value
Single arms			
Smoking status alone	0.794	+0.030	...
Clinical, five variables	0.764	...	...
Model, three seeds	0.618 ± 0.025	-0.108	...
Plus peritumoral branch	0.675 ± 0.017	-0.089	...
Frozen-feature logistic probe	0.617 ± 0.052	-0.147	...
Radiomics	0.578	-0.186	...
Fusion topologies			
Soft voting, model and clinical	0.736 ± 0.005	-0.028	.29-.45
Stacking, cross-fitted	0.718 ± 0.014	-0.046	<.001-.07
Soft voting, equal weights	0.703 ± 0.008	-0.061	.04-.11
Soft voting, clinical and radiomics	0.703	-0.061	.02
Hard voting	0.688 ± 0.006	-0.076	.04-.06
Soft voting, model and radiomics	0.619 ± 0.015	-0.146	.003-.01

Note.—Data are areas under the receiver operating characteristic curve. Values with $\pm$ are across-seed means $\pm$ SDs over the three canonical training runs; the clinical and radiomics arms do not depend on the imaging run and so have no spread. Differences are against the clinical model and are means across runs. Ellipses indicate not applicable. P values are from a paired bootstrap over patients with 2000 resamples and are given as the range across the three runs; P values less than .05 indicated statistical significance. The co-primary endpoint is the soft-voting combination of the clinical and imaging arms — an unweighted average of within-fold ranks — minus the clinical model, computed per run in a separate analysis that re-derives the clinical model inside each run’s own folds; it was -0.029 ± 0.015 , with every run’s CI including zero. AUC = area under the receiver operating characteristic curve, CI = confidence interval.

Table 3: Every External Localization Evaluation, on the Same 420 Held-out NSCLC-Radiomics Patients

Run	Fold	Mass	Shuffle	Lift	Pointing	IoU
A	0	0.375	0.048	+0.327	0.862	0.563
A	1	0.038	0.045	−0.007	0.000	0.001
A	2	0.394	0.048	+0.346	0.855	0.543
A	3	0.038	0.045	−0.007	0.000	0.000
A	4	0.043	0.048	−0.005	0.000	0.000
B	0	0.327	0.049	+0.278	0.833	0.538
B	1	0.316	0.048	+0.268	0.790	0.502
B	2	0.401	0.050	+0.351	0.893	0.570
B	3	0.411	0.051	+0.360	0.831	0.562
B	4	0.550	0.049	+0.501	0.802	0.525
C	0	0.345	0.049	+0.296	0.810	0.524
C	1	0.040	0.044	−0.005	0.000	0.016
C	2	0.306	0.049	+0.258	0.757	0.469
C	3	0.409	0.051	+0.359	0.829	0.550
C	4	0.384	0.048	+0.336	0.874	0.553
Trained, mean		0.292	0.048	+0.244	0.609	0.394
Control: center prior		0.270	0.056	+0.214	0.476	0.287
Control: untrained head		0.049	0.049	−0.000	0.005	0.008

Note.—Runs A, B and C are three independently seeded training runs of five cross-validation folds each. Mass is the fraction of attention falling inside the reference tumor contour, Shuffle is the same values permuted across cells, and Lift is their difference. Pointing is the fraction of patients whose peak attention cell falls inside the contour. Eleven of 15 evaluations exceed their shuffle, meeting the pre-specified threshold of 71.4% by one evaluation. All 11 successes exceed the center prior on pointing accuracy. IoU = area-matched intersection over union, NSCLC = non-small cell lung cancer.